

TORONTO PEARSON, ON, Canada

WMO: 716240

Lat: 43.6658N

Lon: 79.6062W

Elev: 569

StdP: 14.40

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.3	3.8	-11.6	3.0	-0.2	-5.8	4.1	5.1	31.7	22.0	28.5	23.6	11.8	0	0.582

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.1	88.7	72.4	85.5	70.8	82.6	69.4	74.8	84.5	73.0	82.4	71.4	79.6	13.4	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	120.3	80.2	70.0	112.8	78.3	68.4	106.7	76.7	38.7	84.5	37.1	82.8	35.6	79.8	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.6	24.0	21.1	-7.1	93.3	6.3	3.5	-11.7	95.8	-15.4	97.9	-19.0	99.8	-23.6	102.3
			-8.1	78.2	6.1	2.1	-12.4	79.7	-16.0	81.0	-19.4	82.2	-23.9	83.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.0	23.1	24.6	32.9	44.7	56.9	66.9	71.9	70.4	63.3	50.9	39.5	29.8
	DBStd	19.14	11.39	10.11	10.21	8.36	8.20	6.82	5.64	5.41	7.25	7.91	8.25	9.04
	HDD50	3347	836	711	536	201	24	0	0	0	2	85	327	625
	HDD65	6803	1300	1131	994	611	278	57	7	14	115	441	765	1090
	CDD50	2628	0	0	8	40	238	506	677	631	402	114	11	1
	CDD65	610	0	0	0	1	27	113	219	180	65	5	0	0
	CDH74	4501	0	0	1	14	238	858	1726	1201	436	27	0	0
	CDH80	1263	0	0	0	1	51	257	539	302	109	4	0	0
Wind	WSAvg	9.9	11.3	11.2	11.0	10.9	9.4	8.8	8.8	8.3	8.4	9.4	10.1	10.8
Precipitation	PrecAvg	31.7	2.3	2.1	2.1	2.9	2.8	3.1	3.1	2.9	2.6	2.7	2.6	2.4
	PrecMax	40.4	5.1	4.3	4.3	5.1	5.1	6.7	7.0	5.5	6.5	5.2	5.0	4.5
	PrecMin	23.5	0.9	1.1	0.8	1.1	0.4	1.2	0.8	0.7	0.7	0.7	0.8	1.0
	PrecStd	4.6	1.0	0.9	1.0	1.1	1.1	1.6	1.5	1.2	1.2	1.3	1.1	0.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	53.5	67.6	75.7	85.1	90.8	93.4	91.8	88.8	77.7	63.8	55.5
		MCWB	51.2	46.9	56.5	59.9	69.2	73.3	75.0	73.5	72.3	65.3	56.1	50.9
	2%	DB	46.2	45.8	58.8	68.3	80.6	86.7	89.1	86.4	83.0	71.7	59.2	49.1
		MCWB	43.1	41.1	50.6	55.8	65.8	70.8	72.8	71.9	69.9	62.6	53.7	45.6
	5%	DB	41.3	41.0	52.2	63.0	75.7	82.6	85.8	83.1	78.9	67.4	55.7	44.8
		MCWB	39.1	37.4	45.6	51.8	63.3	69.3	70.9	69.7	67.7	59.9	50.8	41.7
	10%	DB	37.7	37.5	46.6	58.1	70.9	78.8	82.5	80.4	75.2	63.2	51.7	41.2
		MCWB	35.3	34.6	41.0	48.5	60.0	66.5	69.5	68.5	65.9	57.1	47.3	38.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.4	47.3	58.8	62.7	71.7	75.5	78.2	76.5	74.4	69.0	58.2	52.0
		MCDB	53.1	51.5	65.8	71.2	82.7	87.3	89.0	84.7	84.3	74.4	61.4	54.3
	2%	WB	43.7	41.9	52.2	58.2	68.2	73.1	75.4	73.9	71.9	63.7	54.7	46.4
		MCDB	45.5	45.2	57.2	66.0	78.1	84.1	84.7	82.9	80.2	69.7	58.8	48.7
	5%	WB	39.3	37.8	46.4	53.6	64.8	70.7	73.2	72.3	69.6	60.7	51.1	42.3
		MCDB	41.1	40.5	51.3	60.9	73.6	79.9	82.4	80.4	76.1	65.9	54.8	44.6
	10%	WB	35.5	34.7	41.5	49.4	61.3	68.5	71.3	70.5	67.2	58.0	47.9	38.5
		MCDB	37.4	37.4	46.0	56.4	69.3	76.5	79.4	77.8	73.4	62.5	51.6	41.0
Mean Daily Temperature Range	5% DB	MDBR	12.7	13.1	14.4	16.5	17.5	16.9	17.1	16.6	16.7	14.4	12.3	10.8
		MCDBR	14.4	16.0	21.8	24.9	22.4	21.2	20.1	19.4	20.7	19.6	17.7	14.0
		MCWBR	13.5	13.8	16.0	16.7	13.2	11.3	9.7	9.4	10.8	12.7	14.1	12.4
	5% WB	MCDBR	14.0	15.4	21.0	21.9	21.0	19.1	17.5	16.9	17.6	18.0	16.7	13.8
		MCWBR	13.9	14.0	16.7	16.9	13.4	11.7	9.7	9.4	10.6	12.8	14.4	13.1
Clear-Sky Solar Irradiance	taub		0.294	0.320	0.331	0.376	0.404	0.434	0.426	0.413	0.381	0.360	0.316	0.298
	taud		2.395	2.310	2.398	2.311	2.283	2.225	2.270	2.296	2.366	2.407	2.490	2.434
	Ebn at Noon		265	275	288	282	276	266	267	267	267	258	256	252
	Edh at Noon		24	31	33	39	41	44	42	39	34	29	22	22
All-Sky Solar Radiation	RadAvg		450	687	1095	1406	1702	1854	1860	1631	1307	783	489	355
	RadStd		44	66	110	150	179	134	134	83	122	72	62	38

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (52 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page